BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 056 · Lesson 056 · Tree Traversals: Post-order and Breadth-first

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a leaf node?

Q2. What is BFS's underlying structure?

Q3. What order does postorder use?

Q4. 022

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

postorder — Left-right-root tree traversal.

BFS — Breadth-first level exploration.

queue — FIFO worklist.

recursive — Process invoking itself on children.

New learning

Understand — Post-order visits both child subtrees before its parent. Traverse left subtree, right subtree, then root.

Model — Breadth-first visits by levels, left-to-right with a queue that stores the next vertices.

Evaluate — Traversal result depends on the precise tree and stated child order. BFS queue changes must be shown to justify an answer.

Worked example

Give postorder for tree root5, left2, right7; node2 has left1.

1. Explain — Apply left-right-root recursively. Do not simply reverse inorder.

We Do · Practise with guidance

Give breadth-first order for root9, left4, right12; 4 has right6.

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Queue starts [9] → enqueue [4,12] → visit4/enqueue6 → visit12 → visit6.
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Explain

Order 9,4,12,6. Visit by depth, using FIFO.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Root8, children3 and10, with 3 children1 and6, 10 right child14. Give postorder. [6]

Question 2

Give breadth-first order for same tree, and first two queue changes. [6]

Question 3

Explain why postorder is useful for deleting tree structures but BFS for shortest unweighted-edge distance. [6]

Exit Ticket · Answer before checking

Breadth-first tree traversal uses a stack to process deepest nodes first.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Tree node with no children.

Q2. A queue.

Q3. Left subtree, right subtree, root.

Q4. BST root10 with left5 and right15; 5 has right7. Give inorder traversal.

You Do pages · Independent questions

Q1. 1 [1],6 [1],3 [1],14 [1],10 [1],8 [1].

Q2. 8,3,10,1,6,14 [4]; start [8] [1] and after 8 [3,10] [1].

Q3. Postorder visits children before parent [2] allowing safe subtree cleanup [1]; BFS explores layers [2] first discovery gives shortest unweighted edge count [1].

Exit Ticket · Model answer

False. Breadth-first ordinarily uses a queue.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: